Behavioural Credit Risk from Repayment Histories

A Comparative Study of Default Prediction

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Independent Research Project

October 2026

1 Abstract

This project asks how much six months of repayment history add beyond an account's latest recorded status, and whether an ordered neural representation is needed to obtain that information. The analysis uses 30,000 historical Taiwan credit-card accounts, with a reused 6,000-account reporting sample containing 1,327 defaults. The follow-up study adds capacity-matched rules, latest-month models, history ablations, five-seed sensitivity and a separate analysis of accounts without a latest recorded delay.

At a fixed 1,200-account review capacity, full-history LightGBM captures 678 defaults (51.09%), compared with 637 (48.00%) for latest-month LightGBM. Its 3.09-percentage-point capture advantage has a paired 95% bootstrap interval of [1.27, 4.63] points. The ordered GRU captures 671 defaults, compared with 648 when aligned monthly vectors are independently permuted within each account. Removing repayment-status channels reduces capture to 559 defaults.

These comparisons support the usefulness of historical information and of consistent order in this fitted GRU. They do not establish that recurrence is necessary: a flattened monthly network captures 665 defaults, and the GRU's capture difference over that model is inconclusive. The GRU also has no clear advantage over LightGBM. Full-history LightGBM is therefore the preferred comparator to advance for the stated review objective.

Within accounts without a latest recorded delay, a dedicated highest-risk fifth captures 37.63% of defaults with the GRU. This corrects an earlier interpretation of a low global-policy recall as proof of general early-warning failure. The findings remain exploratory: the reporting cohort was already examined, and no fresh temporal or external validation has been obtained.

2 Introduction

A credit-risk score can help order existing accounts for review. The useful question is how much information it adds beyond a simple signal that a reviewer already sees. A high overall ranking metric is not sufficient if a model mainly reproduces an existing repayment-status field.

The initial ClearCredit experiment compared statistical, tree and neural approaches. A three-seed GRU was selected using development average precision. It achieved slightly higher reporting AUC than LightGBM, but captured fewer defaults at an exact 20% review capacity. Its paired uncertainty did not establish a clear neural advantage. That result motivated a more focused challenge of what the fitted model had learned.

The follow-up asks three questions. First, does a model improve materially on latest-status rules and latest-month financial inputs at the same workload? Second, does consistent monthly order help a recurrent representation, and how does it compare with the same inputs flattened into a feed-forward model? Third, does low recall under a global policy imply an inability to rank accounts without a latest recorded delay?

The contribution is a controlled retrospective comparison with explicit decision consequences. It adds missing simple benchmarks, distinguishes historical information from a particular architecture, and examines whether the model's complexity is justified. Tree methods remain credible comparators on medium-sized tabular problems [2]. Sequential credit-monitoring research motivates the GRU [3], but does not supply a performance guarantee on these six-month records.

The original development selection is preserved. This follow-up was designed after reviewing the original reporting results, so the additional comparisons are exploratory. A reporting winner is not relabelled as a newly development-selected champion.

3 Data and Evaluation Design

3.1 Population, information and outcome

The UCI Default of Credit Card Clients dataset contains 30,000 existing Taiwan credit-card accounts with histories from April to September 2005 [1]. The binary outcome is the source-defined next-month default-payment label. It is not reinterpreted as a one-year default probability, a standard 90-day delinquency event or a lifetime credit-loss outcome.

The financial inputs are credit limit and six monthly repayment statuses, billed balances and payment amounts: 19 raw variables per account. Sex, age, education, marriage and account ID are excluded from model predictors. Age and sex support descriptive audits; exclusion does not establish fairness.

Negative balances are retained. Status codes -2, -1 and 0 remain distinct in learned categorical representations. No-delay rules use a nonnegative severity score; they do not assign a substantive ordinal interpretation to negative codes. Source metadata report no missing values; later masking is synthetic.

3.2 Partitions and overlap controls

The original stratified random 20% reporting split, seed 42, is retained. After excluding 608 development rows with financial-input profiles present in reporting, a stratified group split separates initial fitting, tuning and calibration accounts. The groups are hashes of the full 19 financial inputs. There are 817 repeated financial rows and 85 groups with conflicting labels; these do not necessarily identify repeated people.

Table 1: Fixed partitions, unchanged from the original experiment.

Partition	Accounts	Use
Initial fitting	14,036	Development candidate fitting
Tuning	4,678	Original selection and latest-model search
Calibration	4,678	Separate sigmoid recalibration
Reporting	6,000	Reused retrospective evaluation

Final base models use the combined 18,714 fitting and tuning accounts. Calibration accounts remain excluded from base fitting. Financial-input groups do not cross the four original partitions. The source checksum and all partition hashes are checked before the follow-up runs.

3.3 What the reporting sample can establish

The reporting sample was already examined in the earlier project. Group separation and bootstrap intervals do not restore an untouched evaluation. The results compare these fitted approaches on a known historical cohort; they do not establish future-cohort stability, transportability to another lender or performance on new applicants.

4 Experimental Methodology

4.1 A common review objective

Every main comparison selects exactly 1,200 of the 6,000 reporting accounts. If $\mathcal{R}$ is the selected queue and y_i the default outcome,

$$\text{Capture}(\mathcal{R}) = \frac{\sum_{i \in \mathcal{R}} y_i}{\sum_{i=1}^{n} y_i}. \tag{1}$$

Queue precision divides captured defaults by reviewed accounts. AUC and average precision assess broader ranking; Brier score and log loss assess probabilities. Capture is the primary decision measure.

A label-independent random tie key is generated once, seed 20261031. Scores are ranked first and ties broken by that key. For tied rule scores, expected capture under uniform tie breaking and sensitivity across 200 additional tie draws are also reported. The tie range is not a confidence interval.

4.2 Rules and latest-month comparators

The binary rule prioritises accounts whose latest status is at least one. The severity rule orders the nonnegative latest status, leaving nonpositive codes tied. Both rules are evaluated at the same exact capacity as learned models. A separate unrestricted binary-policy result reports its naturally larger workload, rather than presenting it as a matched comparison.

Status-only logistic regression receives categorical latest status. Latest-month logistic regression and LightGBM receive credit limit, latest status, latest balance, latest payment and the two latest amount-to-limit ratios. They receive no earlier months or historical summary features. The logistic search uses $C \in \{0.01, 0.1, 1\}$; latest-month LightGBM uses the same three development presets as the original tree comparison. Selection uses tuning average precision, followed by refitting and independent calibration.

Full-history logistic regression and LightGBM retain their original fitted configurations and results. Their engineered features include historical limit ratios, utilisation change, delay counts and payment summaries. Consequently, the latest-versus-history tree comparison assesses the complete added historical feature set, rather than isolating one particular lag or summary.

4.3 Neural history controls

Each monthly vector has four numeric channels, 12 status indicators and one missing/unknown-status flag. Numeric channels are signed-log balance and payment, balance-to-limit and payment-to-limit. A separate transformed credit-limit input is retained. Imputation and scaling are fitted on base-fitting accounts only.

The ordered GRU has hidden width 24, a 16-unit static branch and a 32-unit prediction head. Its representation follows

$$h_{it} = \text{GRU}(x_{it}, h_{i,t-1}). \tag{2}$$

The following controls change the representation while retaining a fixed experimental budget.

The flattened MLP receives exactly the same 102 monthly channels plus the static input, arranged as 103 features. It has two hidden layers of 32 units. It can learn associations with month-specific positions without recurrent state. Its parameter count is close to the GRU's.

The permuted-order GRU receives a fixed, independently chosen month permutation for each account, seed 7331. All channels within a monthly vector move together. The model is fitted, calibrated and evaluated on that representation; this is not merely corrupting inputs to an ordered model at inference. One permutation realisation is used. Financial content can retain chronological clues; this is an order-disruption control.

The no-status GRU removes all 13 status-related channels, retaining four numeric channels and the same static branch. It tests how much the fitted approach depends on repayment-status information.

Table 2: Trainable parameters per single neural fit.

Representation	Parameters
Ordered GRU	4,473
Flattened MLP	4,417
Permuted-order GRU	4,473
No-status GRU	3,537

4.4 Training, seeds and calibration

All neural controls use 26 fixed epochs, dropout 0.2, unweighted binary log loss, AdamW learning rate 0.001 and weight decay 0.001, batch size 256 and gradient clipping at five. These are inherited from the original GRU selection. They supply a controlled budget, not an independently optimised best-case search for every architecture.

The primary ensembles average raw probabilities from seeds 42, 137 and 2026. Seeds 718 and 9001 are additional single-fit sensitivity checks. They do not alter primary ensemble membership. The original fitted models are restored and their reporting scores checked before extension. New candidates are fitted without feedback from their reporting metrics.

A sigmoid recalibrator is fitted on the separate calibration accounts for each ensemble and single-seed evaluation:

$$p_i^{\text{cal}} = \sigma\left(a + b\,\text{logit}(p_i^{\text{raw}})\right). \tag{3}$$

Raw probabilities are clipped before taking logits. Recalibration is assessed as a probability check, rather than assumed to produce a large accuracy gain [4].

4.5 Uncertainty and the practical decision rule

There are 500 paired bootstrap draws of full financial-input groups. Models are compared on the same draws. The intervals condition on fitted models and the reporting sample; they exclude retraining, tuning and previous research choices. Secondary comparisons are not adjusted for multiplicity and remain exploratory. Seed variation is reported separately.

5 Results and Model Challenge

5.1 All comparisons at the same workload

All rows in Table 3 review 1,200 accounts and use the same denominator of 1,327 defaults. Rule rows use the fixed tie key. They are ranking rules, not calibrated probabilities.

Table 3: Retrospective reporting results at exact 20% capacity.

Approach	AUC	AP	Capture	Defaults
Binary delay rule	0.6776	0.3591	44.69%	593
Severity rule	0.6909	0.4252	46.42%	616
Status-only logistic	0.7009	0.4446	46.42%	616
Latest-month logistic	0.7376	0.4903	47.17%	626
Latest-month LightGBM	0.7586	0.5274	48.00%	637
History logistic	0.7647	0.5204	50.04%	664
History LightGBM	0.7753	0.5495	51.09%	678
Ordered GRU	0.7772	0.5521	50.57%	671
Flattened MLP	0.7731	0.5384	50.11%	665
Permuted-order GRU	0.7682	0.5310	48.83%	648
No-status GRU	0.7310	0.4515	42.13%	559

The full-history tree captures 678 defaults, versus 637 for latest-month LightGBM. That is 41 additional defaults at the same workload, an absolute 3.09-point increase in capture. The paired interval is [1.27, 4.63] points. The comparison supports historical features in this fitted tree workflow.

The ordered GRU captures 671 defaults and has AUC 0.7772. Its slightly higher AUC than LightGBM does not translate into more captured defaults. Their AUC difference interval remains $[-0.0031, 0.0071]$. The appropriate conclusion is an inconclusive neural advantage, rather than equivalence or a GRU win.

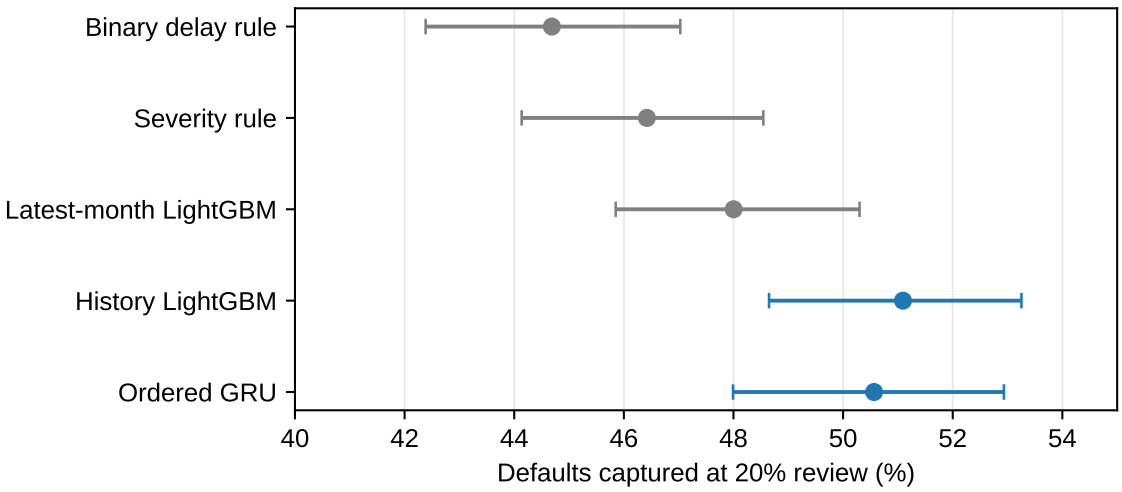


Figure 1: Default capture with 95% paired group-bootstrap marginal intervals. All approaches review exactly 1,200 accounts. Pairwise difference intervals are reported separately.

5.2 How much value is added beyond a simple rule?

The unrestricted binary delay flag reviews 1,324 accounts (22.07%) and captures 660 defaults (49.74%). Comparing that result directly with a 20%-capacity model confounds ranking with workload.

At the common 1,200-account capacity, the fixed-tie binary rule captures 593 defaults (44.69%), and the severity rule captures 616 (46.42%). Their expected capture under uniform tie handling is 45.08% and 46.73%, respectively. Across 200 alternative tie draws, captured-default counts range from 582 to 616 for the binary rule and 606 to 632 for severity. These ranges describe tie sensitivity, not sampling uncertainty.

LightGBM captures 62 more defaults than the fixed-tie severity rule. Its paired capture difference is 4.67 points, with interval [2.94, 6.51]. The status-only logistic model also captures 616 defaults, although its risk ordering and AUC differ from the severity rule. A single latest-status field therefore accounts for substantial concentration, but does not reproduce the full model's matched-workload result.

5.3 What do historical order and status contribute?

The ordered GRU captures 23 more defaults than the permuted-order GRU: a 1.73-point difference, with interval [0.61, 3.07]. Its AUC advantage is 0.0090, with interval [0.0047, 0.0132]. Under this fixed representation and training budget, consistent monthly order helps the recurrent model.

Removing status channels reduces capture from 671 to 559 defaults, an 8.44-point loss. The paired capture interval is [6.43, 10.23] points. Financial amounts still contain predictive information, but recorded repayment status is an important component of the fitted GRU.

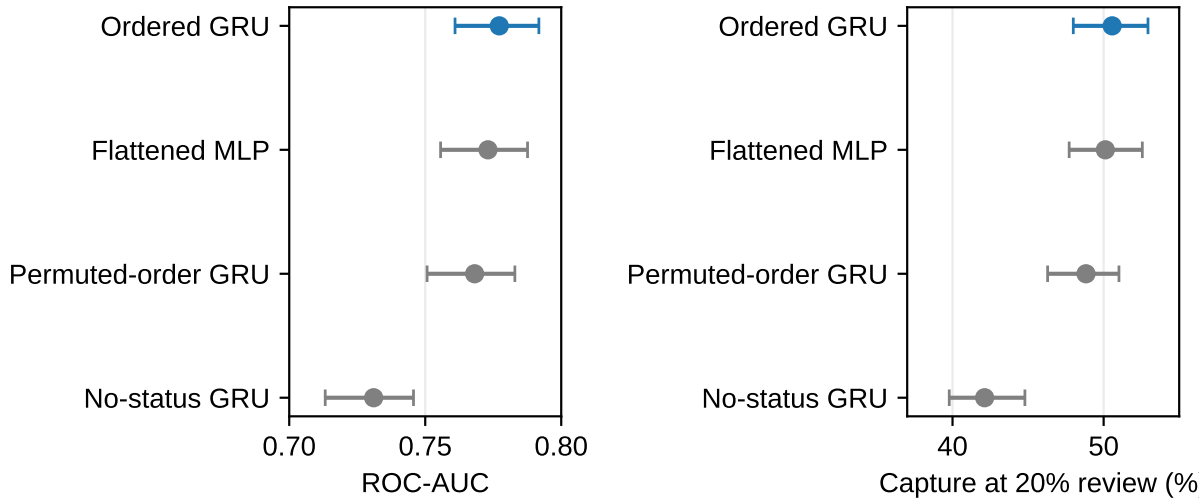


Figure 2: Neural representation controls with 95% marginal bootstrap intervals. All primary ensembles contain the same three seeds. Left: ROC-AUC. Right: default capture at 20% capacity.

The flattened MLP captures 665 defaults, only six fewer than the ordered GRU. The capture difference is 0.45 points, with interval $[-0.44, 1.15]$. Although the GRU's AUC and AP contrasts

are positive in these exploratory intervals, recurrence has no clear advantage over flattening for the primary review-capacity objective. The MLP is not independently tuned to its best possible configuration, so this is not a definitive architecture ranking.

Table 4: Observed capture differences and paired 95% intervals, in percentage points.

Comparison	Difference	Interval
History tree minus latest-month tree	+3.09	[1.27, 4.63]
Ordered minus permuted GRU	+1.73	[0.61, 3.07]
Ordered GRU minus flattened MLP	+0.45	[-0.44, 1.15]
Ordered GRU minus history tree	-0.53	[-2.05, 1.03]

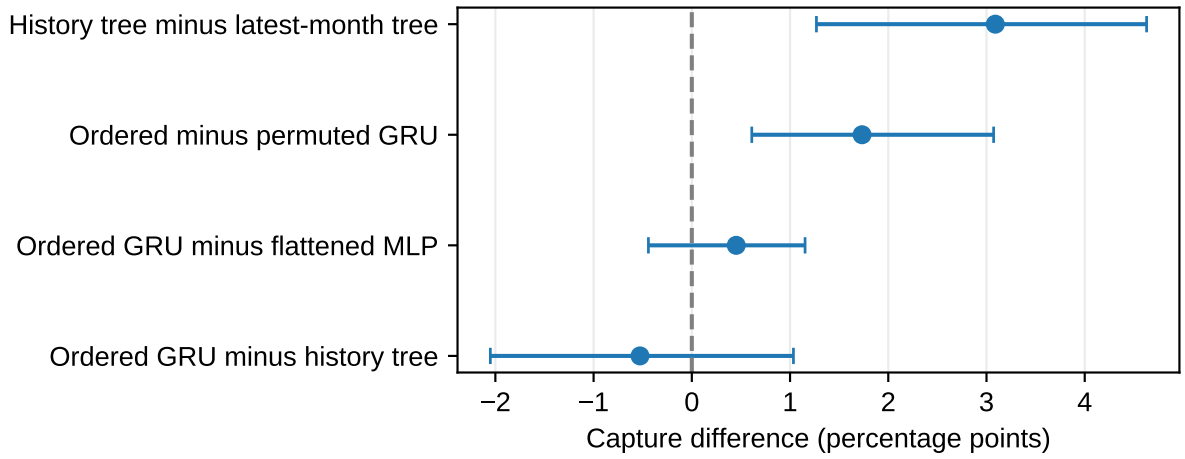


Figure 3: Paired capture differences. The zero line indicates no difference. These conditional exploratory intervals do not account for previous researcher choices or retraining.

5.4 Variation across training seeds

Table 5 reports five individual fits for each neural representation. Each individual score is independently recalibrated on the same calibration partition. The primary ensemble remains the original three-seed average; extra seeds are not selected based on reporting performance.

Table 5: Single-fit ranges across five seeds; capture values are percentages.

Representation	AUC range	Capture range	Ensemble capture
Ordered GRU	0.7745–0.7778	50.41–50.79	50.57
Flattened MLP	0.7679–0.7731	49.66–50.87	50.11
Permuted-order GRU	0.7640–0.7690	48.38–48.76	48.83
No-status GRU	0.7292–0.7317	42.13–42.65	42.13

Ordered-GRU capture ranges from 50.41% to 50.79%. Flattened-MLP capture ranges from 49.66% to 50.87%. The overlap cautions against treating small point differences as robust evidence of architectural superiority. A single MLP seed exceeds the ensemble GRU's capture, but is not promoted after reporting inspection.

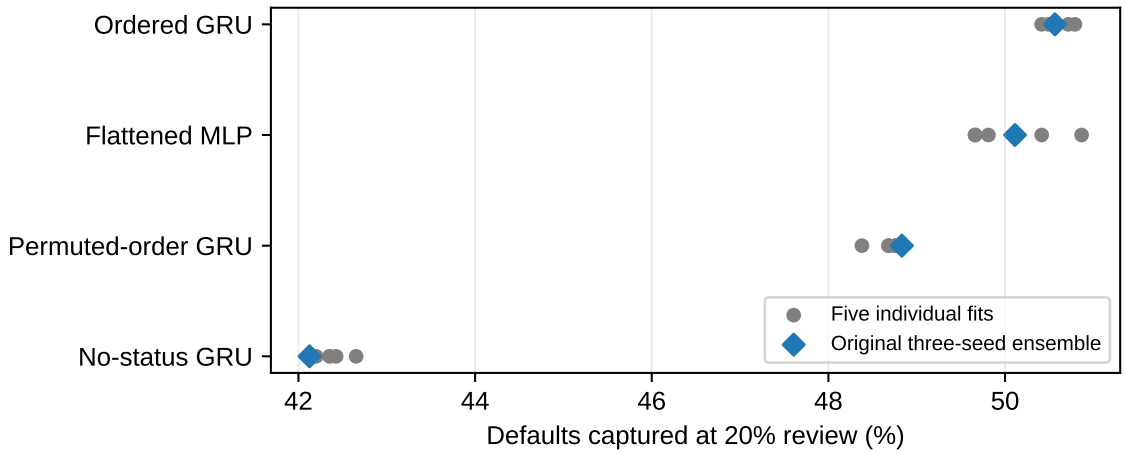


Figure 4: Five individual fits per representation and the original three-seed ensemble. Points describe training-seed sensitivity, not independent datasets or confidence intervals.

6 Account Review and Model Choice

6.1 Correcting the early-warning interpretation

An earlier result found that the original global GRU policy missed 85.2% of defaults among accounts with latest status below one. That is conditional recall under one global threshold. It does not by itself establish an inability to rank accounts within that population. A global queue can devote most of its capacity to accounts with higher visible risk.

There are 4,676 reporting accounts with latest status below one and 667 defaults. Of these accounts, 641 have a delay code in an earlier month. This group therefore means no latest recorded delay, not no previous distress or a verified first-deterioration population.

Table 6 creates a separate highest-risk fifth within this group: 936 accounts, because capacity is rounded up. It changes the review allocation and is not the same overall policy as the global 1,200-account queue.

Table 6: Within-group ranking without a latest recorded delay. Each queue contains 936 accounts; capture denominator is 667 defaults.

Model	AUC	AP	Capture	Defaults
Status-only logistic	0.5235	0.1511	24.29%	162
Latest-month LightGBM	0.6511	0.2319	35.08%	234
History LightGBM	0.6789	0.2670	37.93%	253
Ordered GRU	0.6832	0.2742	37.63%	251
Flattened MLP	0.6750	0.2654	37.03%	247

The GRU captures 251 defaults (37.63%), compared with 253 for full-history LightGBM and 234 for latest-month LightGBM. The binary and nonnegative-severity rules have constant scores in this group, giving AUC 0.5. The learned models retain within-group ranking information. This does not prove genuine early-warning usefulness, which requires a separately defined future deterioration outcome and independent later data.

6.2 A common fixed-threshold diagnostic

The follow-up also fixes each model's threshold at its calibration-score 80th percentile. This common policy differs from the original tuning-score threshold. It is used only to illustrate how a fixed global threshold allocates review capacity, while the main model comparison remains exact-capacity ranking.

Table 7: Global fixed-calibration policy and capture among accounts without a latest delay.

Model	Global workload	Within-group capture
History LightGBM	19.18%	12.14%
Ordered GRU	19.05%	12.74%

Low subgroup recall under this global policy and higher recall under a dedicated subgroup queue can both be true. They answer different allocation questions. Neither result measures whether reviewing an account prevents its default.

6.3 Which model should advance?

For the specified exact 20% review objective, full-history LightGBM is the preferred candidate for further validation. It captures seven more defaults than the original development-selected GRU, while the paired interval does not establish a GRU advantage. This is a retrospective research recommendation, not a claim that LightGBM is a newly selected untouched-test winner or is ready for deployment.

For this review-stage decision, added GRU complexity requires at least a two-percentage-point absolute capture advantage over full-history LightGBM, a positive lower paired 95% bound, and median inference time no more than twice LightGBM. This is a declared research criterion, not a validated monetary threshold. It does not determine deployment approval.

The two-point minimum capture improvement criterion is not met. There is no reason to retain the GRU as the operational preference merely because it is more elaborate. The neural controls remain valuable research evidence about the representation.

Table 8: Median raw-score inference time for 6,000 accounts, seven repeated calls after warm-up on this CPU environment. Preprocessing is included; calibration is excluded for every model.

Model	Milliseconds
Status-only logistic	1.6
Latest-month LightGBM	43.1
History LightGBM	56.0
Ordered GRU	37.6
Flattened MLP	7.4

The GRU is faster than LightGBM in this particular batch measurement. The experiment therefore does not support a claim that neural inference is necessarily slower. Its added complexity is not justified by the primary predictive result; latency is not the failing condition. Batch timing does not measure service latency, concurrency, monitoring or maintenance cost.

7 Probability and Input Diagnostics

7.1 Calibration and the size of the gain

The original calibrated GRU predicts a mean default rate of 21.56%, compared with 22.12% observed. Its reporting diagnostic slope is 0.987. They do not establish accurate probabilities in every group. The follow-up restores the original predictions and preserves those measured values.

Table 9: Probability errors on the same reporting accounts. Lower is better.

Model	Brier score	Log loss
History LightGBM	0.13576	0.43223
Ordered GRU	0.13551	0.43119
Flattened MLP	0.13645	0.43384
Latest-month LightGBM	0.13917	0.44182

Sigmoid calibration changes the GRU's Brier score only from 0.13558 to 0.13551. It should be described as a probability-reliability check, not a major predictive improvement. The fifth original GRU risk decile has 13.17% mean predicted risk versus 15.67% observed, illustrating a local gap despite close aggregate values.

7.2 Sensitivity, subgroup differences and prevalence

Original-policy diagnostics are retained as supporting evidence, rather than silently recomputed under a different threshold. Masking the latest status, balance and payment changes 18.05% of original GRU review flags. Median absolute probability movement is 9.47 points and the 95th percentile is 24.19 points. This synthetic masking does not validate a response to naturally missing history. A missing-input fallback remains untested.

The original status-group permutation reduces AP by about 0.275, versus 0.037 for balances, 0.044 for payments and 0.030 for credit limit. Such permutations may create unrealistic combinations and describe predictive dependence, not causal effects. The new no-status refit provides a complementary comparison by allowing the model to learn again without those channels.

Original age audits show 22.29% predicted risk versus 24.49% observed for accounts aged at least 45, and 23.28% versus 22.33% below 30. These descriptive gaps warrant investigation but do not prove a cause or justify group-specific decisions.

An original pure-label-shift scenario raises default prevalence to 35% while holding input distributions conditional on outcome unchanged. Weighted mean prediction is 24.33% before prior-odds adjustment and 34.64% after it. This illustrates a population assumption; it is not a recession forecast, and the true future prevalence would need estimation. These supporting diagnostics do not measure realised savings or prevented defaults.

8 Limitations and the Next Independent Study

The main limitation is unchanged: these are reused historical outcomes from one lender population and one observation cohort. The post-review design and its unadjusted secondary intervals support exploratory conclusions. They do not justify confirmatory significance claims or broad architecture superiority.

The tree history comparison combines earlier inputs with engineered summaries. Neural controls have a shared fixed budget inherited from the selected GRU, rather than independently optimal configurations. One within-account order permutation is tested. Five seeds describe optimisation sensitivity, not uncertainty across economic periods. Timings are local batch measurements.

A separate later study needs legally usable account histories with real observation dates, stable account keys and fully matured outcomes. The accompanying next-study protocol specifies a customer-disjoint chronological benchmark, development-only rolling tuning, a later calibration window and an untouched reporting window. Arbitrary IDs and the six predictor months cannot manufacture such validation from the present cohort.

That study should retain the simple rules, latest-month controls, full-history tree, flattened network and ordered/permuted GRU comparison. It should freeze the workload, practical gain criterion and subgroup outcome before evaluation. If the neural criterion fails, retain the negative finding. Independent challenge, policy economics and operational approval remain separate requirements.

No exposure, recovery or intervention data are available. A concentrated review queue does not show that review prevents losses. No production deployment or new external validation has been completed as part of this follow-up.

9 Conclusion

The added experiments answer a more precise question than the original model leaderboard. Historical financial information improves this retrospective tree workflow beyond latest-month inputs. Consistent order also improves the tested GRU over a within-account permuted representation, while repayment-status channels contribute substantially to capture.

These results do not establish a need for recurrence. The ordered GRU has no clear capture advantage over a flattened network or full-history LightGBM. At the stated workload, LightGBM captures 678 of 1,327 defaults and is the preferred candidate to advance for independent validation.

The subgroup analysis also corrects an overstatement: low recall under a global threshold does not prove an inability to rank accounts without a latest delay. A dedicated subgroup queue captures 37.63% of those defaults with the GRU, although genuine early-warning value remains unestablished.

The project now provides capacity-matched baselines, targeted ablations, seed sensitivity, paired uncertainty and an explicit model recommendation. Its next substantive milestone is independent later-outcome validation, not a larger network.

10 Reproducibility and References

Evidence and implementation

Repository: https://github.com/Fakhruddin12/clearcredit.

The original study is preserved in `research/v2`. The follow-up is in `research/v3/challenge.py`; its protocol, data manifest, environment, development search and aggregate results are saved alongside it. This report uses the follow-up executed on 31 October 2026. Original models were restored and their results checked before extension. New ablations and latest-month models were fitted and evaluated in the recorded run.

The original source snapshot is commit:

`99f3fbb17c352f1bc2192e6647d8431cf00a7440`

The follow-up source SHA-256 begins `e4a40946fef9fea0`; its full value is recorded in the implementation notes. The workbook checksum and partition hashes match the original manifest. Five-seed predictions and fitted parameters are generated locally. Only permitted summary evidence is committed.

From the repository root, install the pinned requirements, then run:

```
cd research/v3
python challenge.py --data ../v2/data/source.xls \
  --out results_reproduced
python -m pytest -q tests ../v2/tests
python verify_challenge.py \
  --data ../v2/data/source.xls \
  --results results_reproduced
```

If baseline fitted artifacts are absent, the challenge runner reproduces the original experiment first. The restore verifier checks all eleven candidate/rule score vectors on 6,000 reporting accounts, then recomputes aggregate and subgroup metrics. Sixteen data-handling and numerical tests pass. The next independent study is specified separately and has not been run.

Selected primary sources

[1] Yeh, I. (2009). Default of Credit Card Clients. UCI Machine Learning Repository. DOI: 10.24432/C55S3H. Dataset licence: CC BY 4.0.

[2] Grinsztajn, L., Oyallon, E. and Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on tabular data? NeurIPS. https://arxiv.org/abs/2207.08815.

[3] Clements, J. M., Xu, D., Yousefi, N. and Efimov, D. (2020). Sequential Deep Learning for Credit Risk Monitoring with Tabular Financial Data. https://arxiv.org/abs/2012.15330.

[4] Guo, C., Pleiss, G., Sun, Y. and Weinberger, K. Q. (2017). On Calibration of Modern Neural Networks. ICML, PMLR 70, 1321–1330. https://proceedings.mlr.press/v70/guo17a.html.